E-PSS: the Extended Polarimetric Slope Sensing technique for measuring ocean surface waves

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***Abstract*—Polarimetric slope sensing (PSS) is a powerful tool for remote measurement of the spatiotemporal characteristics of short surface waves. However, two key shortcomings have negatively impacted the widespread adoption of the technique: (1) the quality of the measurement may be sensitive to variability in environmental illumination conditions and (2) waves longer than the imager field of view are not reliably measured. We present an extension of PSS– namely, the Extended Polarimetric Slope Sensing (E-PSS) technique, which is robust to variation in ambient illumination conditions and provides simultaneous access to the directional spectrum for both long and short waves from a single surface-looking, narrow FOV camera.**

***Index Terms*—Polarimetry, ocean waves, wave slope sensing, polarimetric slope sensing, wave spectra**

I. Background

THERE is a pressing need to measure the spatiotemporal characteristics of short-scale surface waves. This requirement is primarily driven by the intimate connection between these intermediate to small-scale waves and the local wind forcing: surface waves with wavelengths between 1 cm to 10 m support a significant fraction of the wave form drag, making them an essential mediator of air-sea momentum flux [1]. It is fortuitous that radar-based remote sensing techniques operate with electromagnetic wavelengths in this very range; specifically, from HF radar down to k_u band microwave scatterometry [2]. This enables indirect measurement of wave height, surface roughness, and wind forcing through interpretation of signatures in the backscattered power [3].

However, the most commonly available techniques for measuring the surface wave field are unable to adequately characterize the energy or directionality of the surface wave field at scales shorter than 10 meters. Surface buoys are by far the most ubiquitous of *in situ* wave sensors, but even the small form-factor Spotter buoys do not provide reliable directional information beyond frequencies of 0.8 Hz (roughly corresponding to surface wave scales of 2 to 3 meters) [4]. This leaves a great deal of the surface wave spectrum which remains broadly inaccessible to observation. There are several techniques which have been developed to push the resolution of directional wave measurements to smaller and smaller scales. Perhaps the most widely adopted of these is stereophotogrammetry [5], [6]. This technique is able to recover the spatiotemporal characteristics of surface waves ranging in scale from 100 meters down to 50 centimeters. While this is quite impressive, it still does leave unresolved the shortest gravity and gravity capillary waves. Polarimetric slope sensing (PSS) was developed by Zappa and colleagues in 2008 [7] following decades of development of the application of classical optical theory towards inferring the shape of a free surface from the polarization state of light reflected off of it [8]. PSS exploits the known (or at least, modeled) relationship between the polarization state of light reflected off an air-water interface and the geometry of the reflecting facet. When repeated spatially (over an imager field of view) and temporally (in short succession, as with video), one is able to recover the spatiotemporal evolution of the wave field down to very fine scales. Nevertheless– this technique is susceptible to variation in illumination conditions [7]. And as initially presented, PSS could reliably measure the slopes of surface waves with wavelengths ranging from a few millimeters up to one meter (but in practice, rarely larger) [9].

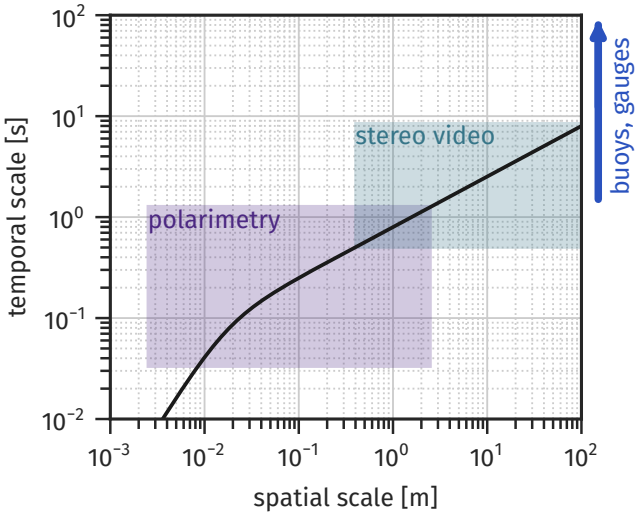


Fig. 1. Stommel diagram showing the spatiotemporal range of select wave measurement techniques: polarimetric slope sensing, stereo video, and point-like objects (buoys, fixed gauges). The black curve represents the gravity-capillary linear dispersion relation. Note that the point-like objects provide no direct access to spatial information.

The Stommel diagram shown in Fig. 1 compares the spatiotemporal ranges of select wave measurement techniques: PSS, stereo-video, and point-like gauges. In order to obtain a true spatiotemporal measurement of the surface wave field,

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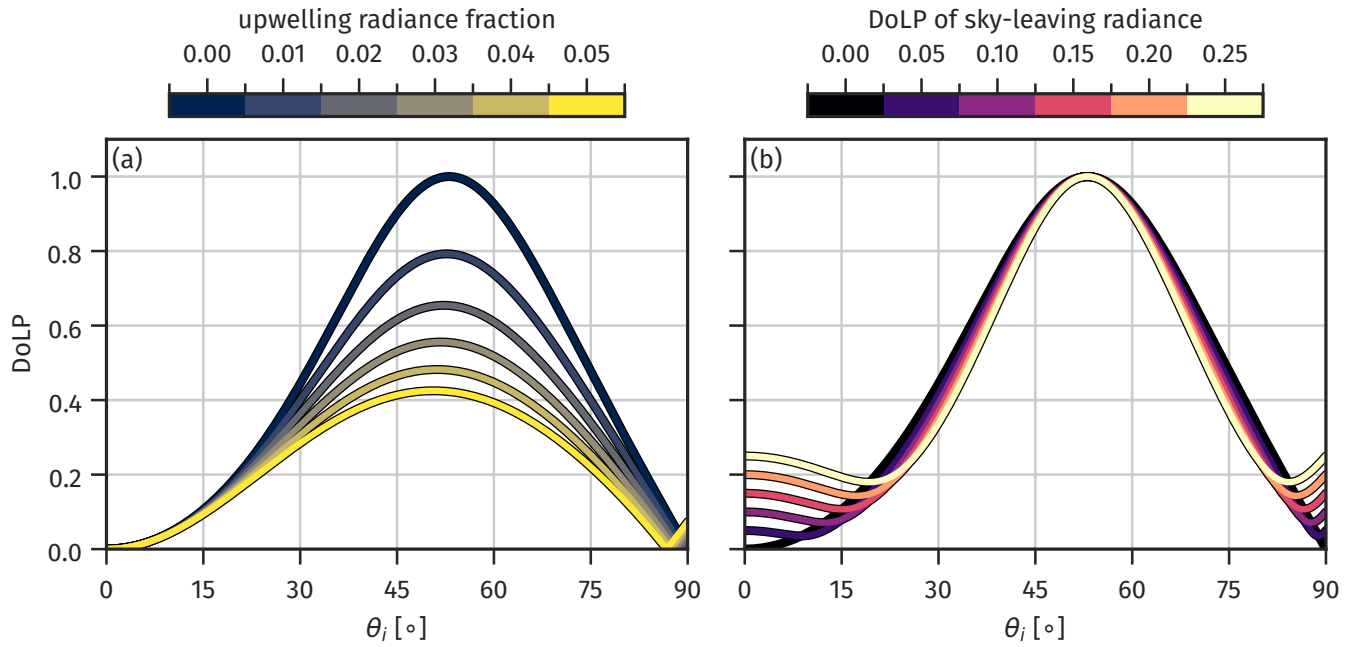


Fig. 2. Fresnel curves, specifying the relationship between incidence angle (θ_i) and degree of linear polarization (DoLP) given variations in (a) fraction of received radiance which is upwelling and (b) DoLP of sky-leaving radiance.

it is important for the observational technique to envelop the gravity-capillary dispersion relation, here provided in the black curve on Fig. 1. The violet shaded region corresponding to PSS is somewhat flexible: by increasing the imaging field of view (i.e., by increasing the freeboard above the mean sea surface), one may shift the region to the right, therefore extending the effective maximally-resolvable temporal scale. Practitioners of wave-sensing stereophotogrammetry have extended the effective temporal window of that technique through time series analysis; e.g., Benetazzo *et al.* [10] employed the maximum entropy method (MEM) technique [11] to their surface elevation and slope estimates to infer the directional spectra of waves larger than the stereo field of view.

II. The Long and Short of It

A core assumption baked into the single-camera application of PSS is that sky-leaving radiance is unpolarized and upwelling (surface-leaving, but scattered from below the interface) radiance is negligible [7]. This allows one to constrain the relationship between a measurement of degree of linear polarization (DoLP) and the incidence angle (θ_i) of the light ray. In practice, however, even small variations in ambient environmental conditions can have a significant impact on the relationship between the measured degree of linear polarization and the incidence angle that one might infer [12], [13]. In Fig. 2(a), we show that even a small fraction of radiance which is upwelling and unpolarized can have a significant impact in compressing the DoLP that would be measured by an imaging polarimeter. Conversely, as shown in Fig. 2(b), the effect of polarized sky-leaving radiance is to raise the floor of the variation of DoLP across all values of θ_i. The combined effect of variable sky-leaving and surface-leaving radiance is to modify this relationship. One approach to solving this problem might be to triple the number of imagers, including in one's sensing suite a floating polarimeter for obtaining upwelling light measurements and a sky-oriented polarimeter for obtaining downwelling light measurements [14]. This would allow one to incorporate these measured values into the Fresnel reflection coefficients and directly compute the relationship between DoLP and θ_i following Mueller calculus. However, one of the greatest benefits of polarimetric slope sensing is that it requires a single camera with no in-water component and no sky-looking camera. Adding additional requirements for in-water recovery of light polarization and sky-looking state would render the technique somewhat esoteric and limit it to specialists in the field of environmental optics. Another approach might be to have two cameras but rather than have one look at the sky and one at the surface we make a measurement of our slope field with a narrow field of view lens and have a second camera alongside it to make a direct measurement of the relationship between degree of linear polarization and incidence angle with a wide format lens. This direct measurement of DoLP(θ_i) would natively account for the sky leaving, the polarization state of sky leaving radiance and the fraction of light which comes up through the surfaced unpolarized. Furthermore, it would also implicitly account for path losses to the polarization state, making it an attractive option for airborne applications of polarimetric slope sensing. We will address the viability of that two-camera approach (and other sophisticated techniques, such as "PSSm" [15]) later in the manuscript. For now, we focus on attempting to solve our problem with a single downward-looking, narrow field of view camera, setting the task before us to correct for unpolarized upwelling radiance and polarized sky leaving radiance in a way that is robust, flexible, usable on already-collected datasets, and does not depend on having multiple in water and sky looking imaging polarimeters.

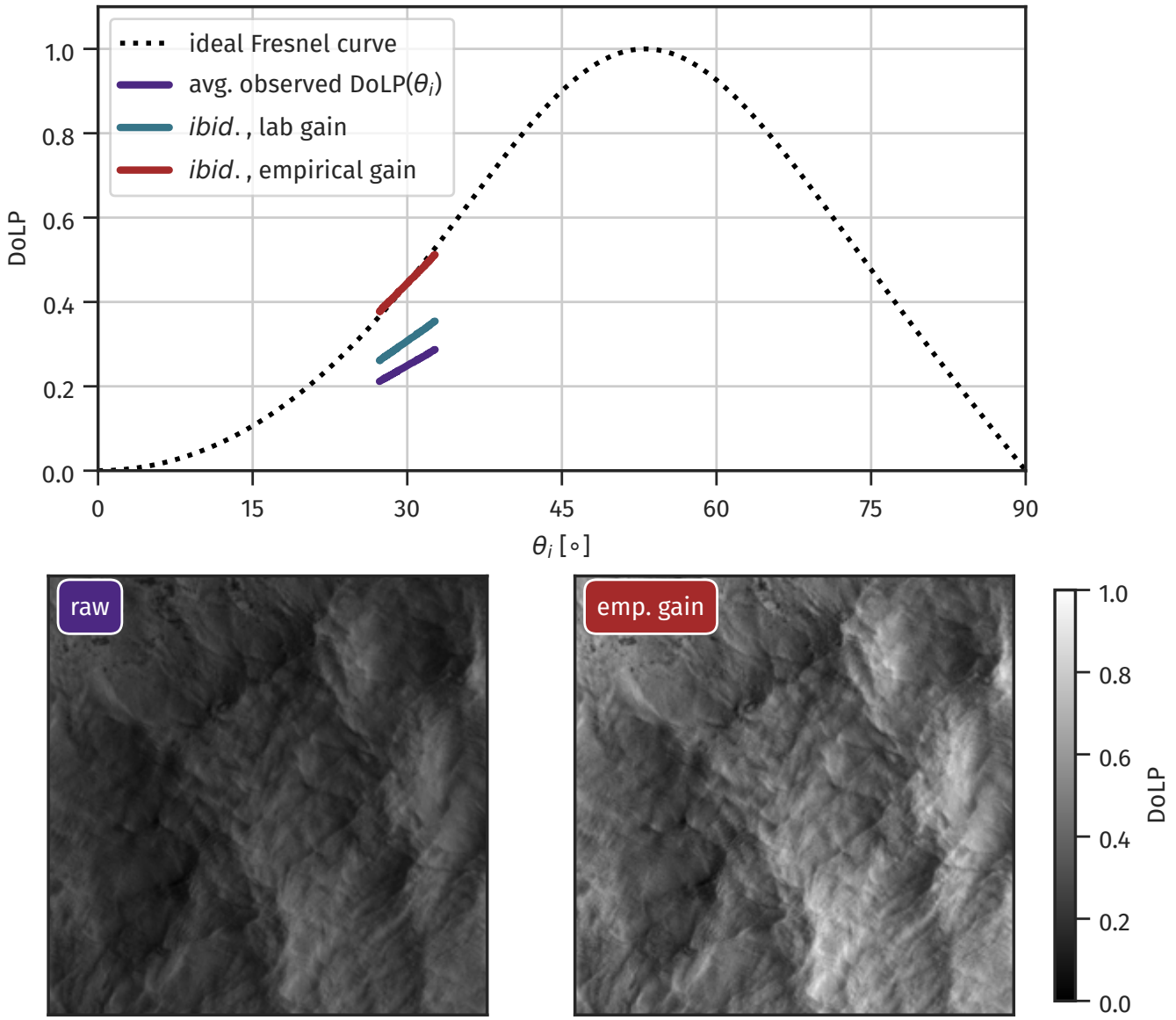


Fig. 3. Top: variation of degree of linear polarization (DoLP) with incidence angle (θ_i) as computed through Mueller calculus or observed via polarimetric camera. Bottom: spatial variation of DoLP provided no gain or an empirical gain; both panels share the same scale and colorbar.

Even under pristine laboratory conditions, a polarimeter may not perform ideally (that is, show ideal response when subjected to an integrating sphere/rotating linear polarizer test). This may be due to a number of factors, among them depolarization of light by optics or deviation of the sensor from optimal sensitivity. Correcting for these persistent effects is essential to making a good measurement [16], [9], [17], [18]. During integrating sphere polarizer tests of our detector, we found the normalized Stokes parameter amplitude to be 0.81 (rather than the ideal 1.0). The difference between S_1 and S_2 amplitudes was determined to not be statistically significant ($p = 0.74$, $N = 12$). We found the mean Stokes parameter to be slightly variable around zero (rms $\approx$ 5%) with a directional asymmetry between minima and maxima of the Stokes parameters on the order of 2-3% of the amplitude. We demonstrate an empirical approach to accounting for variability in upwelling/downwelling radiance in Fig. 3. The black dotted curve in the top panel represents the relationship between DoLP and θ_i (the Fresnel curve) which would arise due to unpolarized sky living radiance and no upwelling radiance. The colored segments correspond to field measurements, with the reported θ_i obtained directly from the known mounting geometry of the instrument and the vertical angle of view for the camera-lens combination (here 30°±2.7° for a 75 mm lens). The violet segment corresponds to the observed variation of DoLP from the bottom to the top of the image frame as averaged over a single ten-minute period. The teal segment represents the same row-averaged DoLP values, but multiplied by the laboratory-obtained gain value of 1.23 (i.e., 1/0.81). The crimson segment corresponds to that set of row-averaged DoLP values subjected to an *empirical* gain, that is: the value by which the frame-median DoLP needs to be scaled in order to yield the value expected for zero upwelling radiance and unpolarized sky-leaving radiance. Corresponding snapshots of the full-frame DoLP in the raw state and post-empirical gain are provided in the bottom two panels.

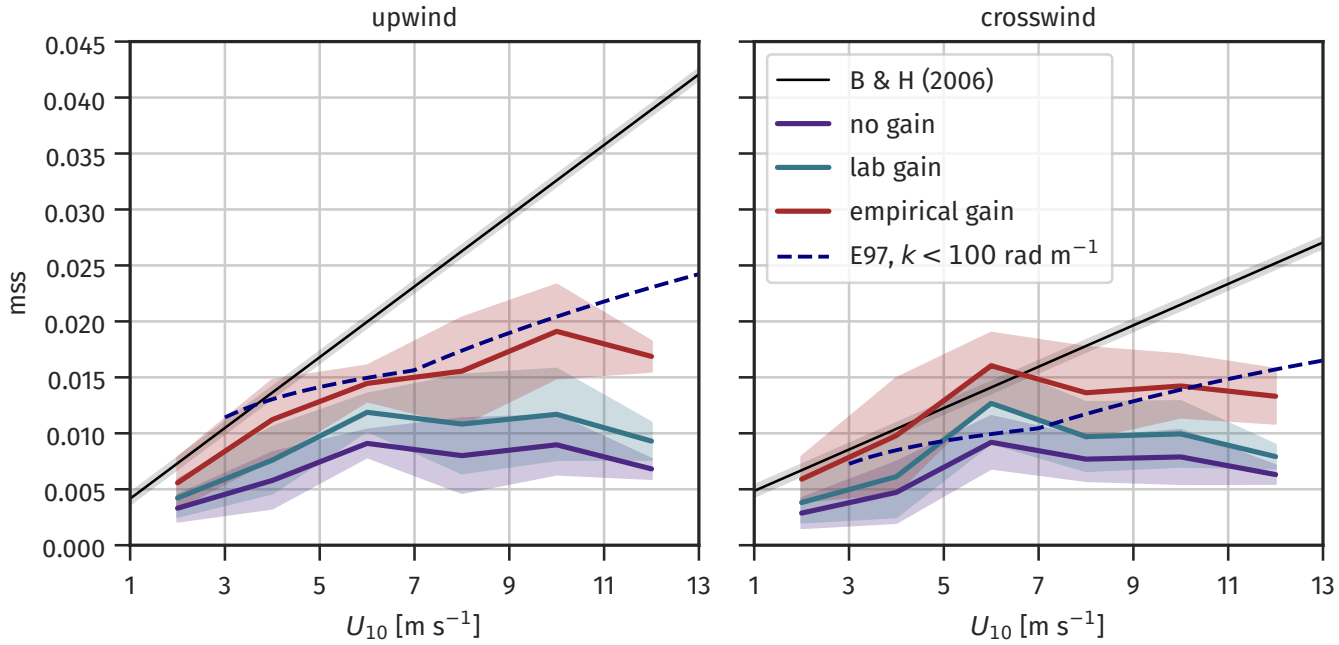


Fig. 4. Upwind and crosswind mean square slope computed from wave slope fields given the three modes of DoLP gain application. Shaded regions mark absolute deviation about the bin median. Black lines and shaded regions represent the least-squares fits and uncertainties of Bréon & Henriot (2006). The dashed blue trace marks values of *mss* computed from the spectrum of Elfouhaily *et al.* (1997) [19] assuming a high wavenumber cutoff of 100 rad m^{-1}.

A. Short Wave Statistics

The dataset we have used to test the efficacy of *post hoc* adjustment to measured DoLP (e.g., as in Fig. 3) is one obtained during a field campaign which spanned the Fall and Winter of 2019-2020. This campaign involved a set of targeted measurements of wind forcing, surface waves, and near-surface currents which were made at the Air-Sea Interaction Tower (ASIT) just south of Martha's Vineyard, MA [18]. As part of this effort (hereinafter "ASIT2019"), we obtained wind speed and turbulent fluxes along with the water surface elevation time series via a triplet of surface-looking near-infrared lidar units. For some observational cases, we also obtained the frequency-directional wave spectrum via bottom-mounted ADCP. More detailed descriptions of this field campaign are readily available in the published literature [20], [21]. The measurements made by our single polarimetric camera pointed at the air-sea interface [18] are the focus of the present work. In Fig. 4, we show the variation of observed wave slope variance (mean square slope) with ten-meter wind speed. The three different colored segments indicate measurements of mean square slope obtained given the three aforementioned DoLP gain modes (no gain, laboratory-derived gain, empirical gain). Shaded regions on the three colored curves from our measurements correspond to 95% confidence intervals about the mean. The black line marks the results of Bréon & Henriot (2006) [22] which are in close agreement with the classic Cox & Munk (1954) [23] observations. The blue dashed trace represents upwind/crosswind slope variance computed from the wavenumber-directional spectrum of Elfouhaily *et al.* (1997) assuming a high wavenumber cutoff of 100 rad m^{-1}. This semi-empirical model [19] was chosen because it reproduces JONSWAP-type spectra [24] at intermediate to long wave scales while upholding the wave slope variance results of Cox & Munk [23] when integrated over all scales. We imposed a high wavenumber cutoff of ≈100 rad m^{-1} to test the hypothesis that the modest values of slope variance relative to Cox & Munk were the result of a scale limitation [18]. In order to provide a more comprehensive description of our observed surface slopes, we computed joint probability density functions (PDFs) in upwind and crosswind wave slope for all ASIT2019 observational cases with the three aforementioned DoLP gain modes. Wind stress and long surface waves impact short waves in a way that nudges their slope distribution to have non-zero skewness and kurtosis (peakedness). Cox & Munk (1954) accounted for this by providing a Gram-Charlier polynomial expansion about a Gaussian distribution [23]. We performed a least-squares fit of the Gram-Charlier expansion of a Gaussian to each wave slope joint PDF. This exercise yielded the coefficients c_{21} & c_{03} (skewness) and c_{40}, c_{04}, & c_{22} (kurtosis). We present these coefficients in Fig. 5, bin-averaged by ten-meter wind speed U_{10} and shown alongside the results of Bréon & Henriot (2006) [22] and Cox & Munk (1954) [23]. Colored segments correspond to the results of E-PSS, with shaded regions representing absolute deviation about the bin median. The shaded regions around the black traces of Bréon & Henriot correspond to their empirically derived levels of uncertainty. We required that the bin-median estimates of c_{ij} were at least twice the 95% confidence interval span for the majority of wind speeds. This threshold was not met for our estimate of c_{03}, hence the gray shading over panels (e) & (f). Our observations of low but indeterminate skewness in wave slope indicates that high-magnitude downwind slopes are being missed. Indeed, even *post hoc* empirical correction is insufficient for recovering information associated with ultra short, very steep waves.

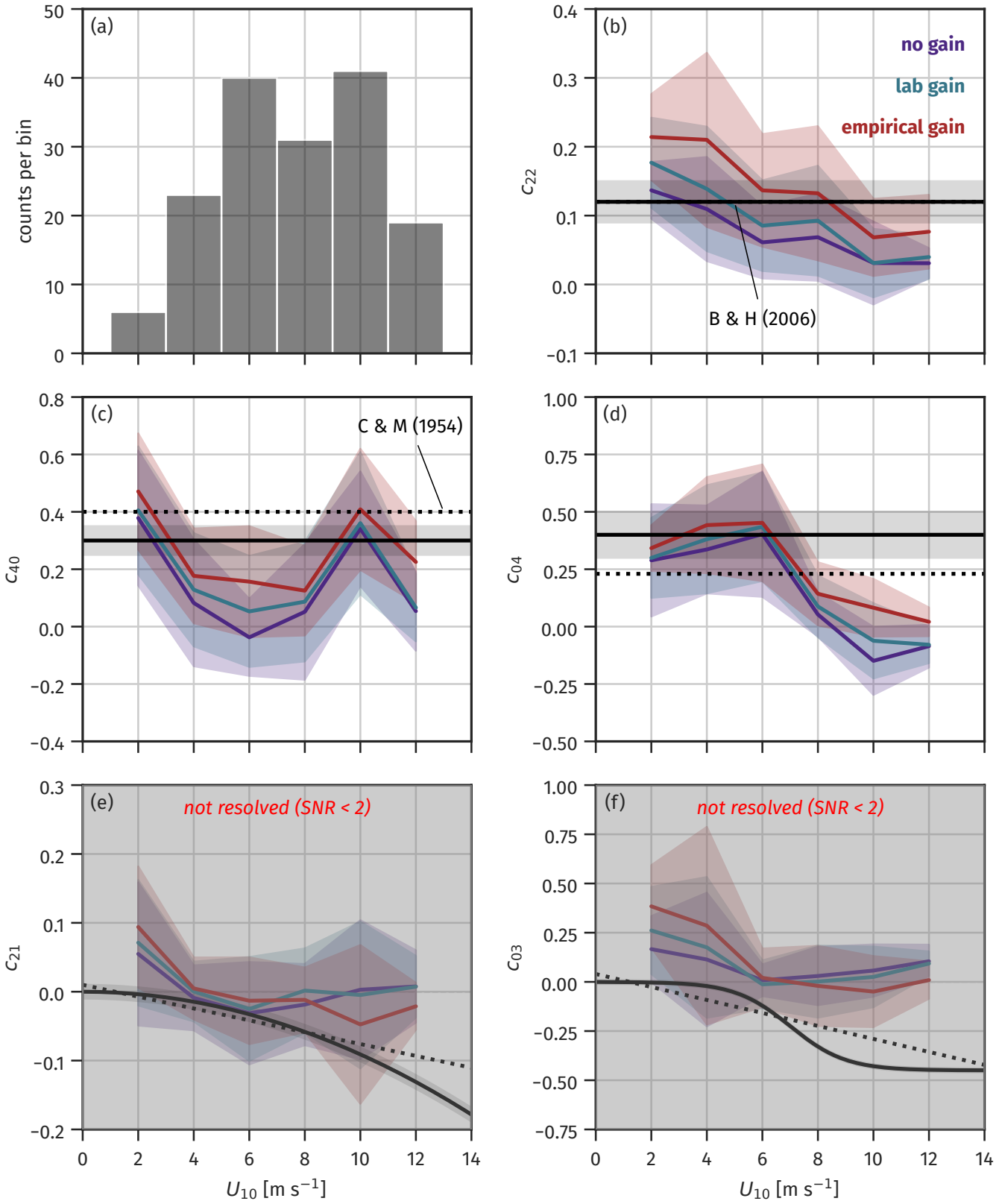


Fig. 5. Coefficients c_{ij} of the Gram-Charlier expansion to the 2D Gaussian wave slope distribution. Coefficients correspond to kurtosis (c_{40}, c_{04}, & c_{22}) and skewness (c_{21} & c_{03}) of the distribution. Colored traces represent values bin-medians equally-spaced in wind speed, with shaded regions marking the median absolute deviation. Black curves represent least-squares fits from Bréon & Henriot (2006), with shaded region marking the qualitatively-determined uncertainty [22]. Bar chart in panel (a) displays sample size per wind speed bin.

B. Long Wave Integral Parameters

As shown in the previous subsection, our single (narrow FOV) surface-looking setup may well be scale-limited on the short end, though there is some evidence that this can be mitigated with a more sophisticated approach [15]. We leave our version of this new approach to the back end of this manuscript. For now, we focus on the other scale limitation: the one imposed by the physical size of the imager field of view. We will attempt to overcome this limit through time-domain analysis: in essence, using the spatially-averaged surface slope time series like a virtual buoy. The simplest way to proceed down this path is to infer the surface elevation spectrum from the slope spectra in Fourier space:

$$F(f) = G(f) \times \Big[\mathcal{S}_E(f) + \mathcal{S}_N(f)\Big] / k^2(f), \qquad (1)$$

where $\mathcal{S}_E(f)$ and $\mathcal{S}_N(f)$ are the frequency slope spectra computed from the (spatially-averaged) time series of eastward and northward slope, respectively, $k(f)$ is the wavenumber inferred via linear dispersion, and $G(f)$ is a (to-be-defined) transfer function. This inversion follows directly from small-amplitude water wave theory, analogous to inversion techniques based on wave orbital velocity/dynamic pressure [25]. There are two practical considerations to keep in mind for this procedure, both having to do with the nature of computing a spatially averaged wave slope time series:

1) A spatially averaged quantity is not a point measurement. It arises due to the convolution between the true quantity of interest and a scale (frequency)-dependent transfer function. This is somewhat analogous to a response amplitude operator (RAO) that one would use to correct for the inertial characteristics of a surface buoy.
2) Small averaging regions ("apertures") provide access to higher frequencies at the cost of contamination by low frequency noise; large apertures beat down low frequency noise at the cost of high frequency information. This is somewhat analogous to the tradeoff associated with airborne imagery at varying altitudes (i.e., optimizing for resolution vs. field of view).

The transfer function associated with the spatial average over a disc of diameter D is $G(f) \equiv 1/H(f)$, where $H(f)$ is the *jinc* function, the radial analogue to the *sinc* function [26]. As with an RAO, $H(f)$ varies between 1 (at low frequencies) and 0 (at high frequencies). Properly, the transfer function goes like $H(k) = 2\frac{\mathcal{J}_1(kR)}{kR}$, with $H(0) = 1$, so we must again invoke small amplitude wave theory to obtain $k(f)$ and produce $H(f)$ which can be used to adjust our frequency slope spectra. Note that $\mathcal{J}_1$ is a Bessel function of the first kind. The second consideration (i.e., the "aperture" problem) leads to a real conundrum: should one we optimize for low or high frequencies when selecting the averaging diameter? Small discs resolve high frequency motions at the cost of amplifying low frequency noise; large discs tamp down low frequency noise at the cost of obscuring high frequency motions. We opted for a disc of diameter $D = 1.46$ m– half the total frame size. This provided good response down to $f \approx 0.08$ Hz without requiring too much amplification from the *jinc* correction at intermediate scales ($f \approx 0.5$ Hz).

The omnidirectional frequency spectra which result from this Fourier-space conversion are provided in Fig. 6. There we see that for this case, the spectra produced via (1) capture the shape determined via lidar. We also see that the DoLP-level gain has a substantial effect on the spectral energy density across all scales, with the empirical DoLP gain yielding a spectrum closest to the lidar spectrum for this case. We have faded the spectral energy densities outside of the aforementioned range 0.08 Hz $< f <$ 0.5 Hz. At frequencies below 0.08 Hz, noise is boosted in the E-PSS spectra (as k^2). At frequencies above 0.5 Hz, the *jinc* correction also boosts under-resolved waves. Furthermore, though the lidar was sampled at 1000 Hz and time-averaged to 10 Hz, its *range accuracy* of $\approx$ 1 cm limited its ability to adequately resolve waves with $f >$ 1 Hz.

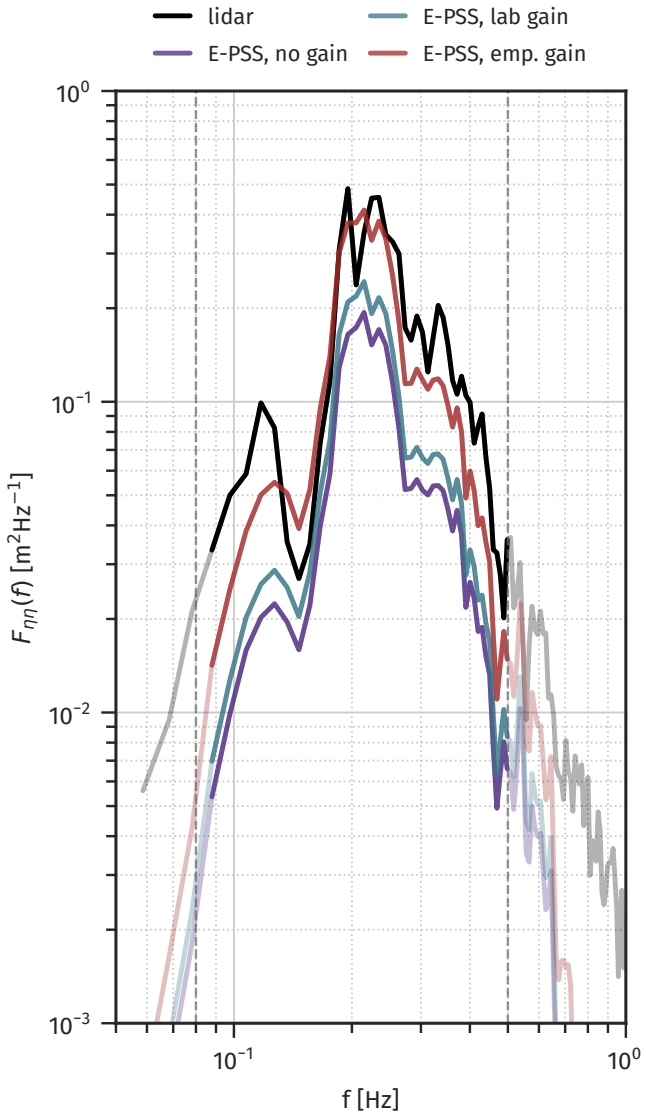


Fig. 6. Omnidirectional spectra computed from water surface elevation timeseries obtained by LiDAR and E-PSS.

A more systematic comparison may be made by comparing integral measures of the spectra, in particular the significant wave height and the peak wave period. We compute the significant wave height as four times the square root of the zeroth moment of the wave spectrum, $H_{m0} = 4 \times \left(\int_0^\infty F(f)\ \mathrm{d}f \right)^{1/2}$.

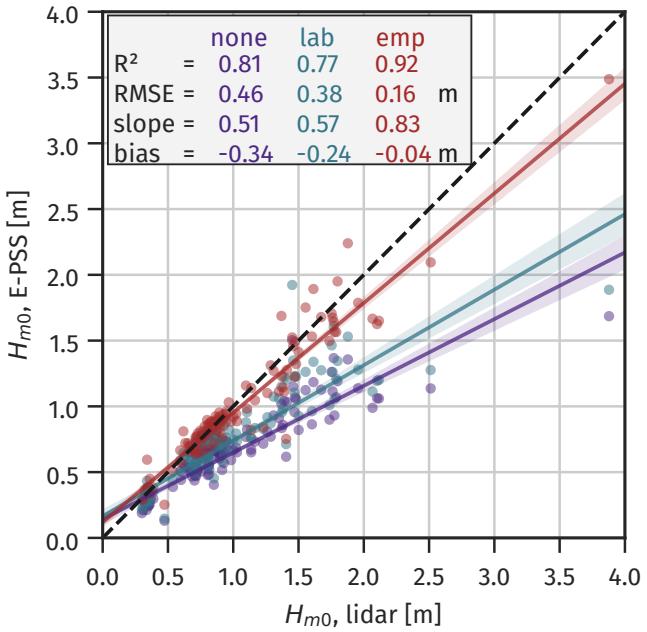


Fig. 7. Scatterplot comparing significant wave height (H_{m0}) estimates from lidar and E-PSS for each of the three DoLP gain values. Dots mark individual observations while lines represent maximum-likelihood fits with uncertainty in both variables [27]; shaded regions are 95% confidence bands on each fit.

We do not compute the peak period via the reciprocal of the discrete spectral peak frequency, as this will depend strongly on whatever single frequency bin happens to have the highest energy. Rather, we adopt the weighted-integral estimator of Young (1995) [28]:

$$T_p \equiv \frac{\int_0^\infty F(f)^q\ \mathrm{d}f}{\int_0^\infty f\, F(f)^q\ \mathrm{d}f}, \tag{2}$$

with $q = 4$, following Young's recommendation. Raising the spectrum to the fourth power concentrates the integral near the spectral maximum while leaving T_p a continuous function of the spectrum. Scatterplots comparing lidar-derived and E-PSS-derived H_{m0} and T_p are provided in Figs. 7 and 8, respectively. The trio of lidar units (three Riegl LD90-3, each at a vertex of an equilateral triangle with edge length of 1.6 m) allowed for an enhanced level of reference quality control: we computed per-record lidar uncertainty from the disagreement between the individual measurements of the same sampled water surface. This uncertainty was found to be small for H_{m0} (of order 1%) and larger for T_p. A record in which one unit's H_{m0} departed from the across-unit median by more than 10% was rejected (two records were excluded on this basis, leaving $N = 112$). After that initial quality control, both H_{m0} and T_p comparisons were fit by maximum likelihood allowing for uncertainty in the lidar reference as well as in E-PSS using the unified equations of York *et al.* (2004) [27]. For uncorrelated errors this is equivalent to the effective-variance approach of Orear (1982) [29]. Ordinary least squares treats the reference as exact, which attenuates the fit slope in proportion to the reference's own noise. The per-record lidar uncertainty is taken from the inter-instrument disagreement, with the (*a priori* unknown) E-PSS uncertainty scaled by a single constant until the reduced $\chi^2 = 1$. The E-PSS uncertainty scaling factor is 21 for H_{m0} and 8 for T_p, confirming that the reference is considerably more precise than the residual disagreement between the lidar array and E-PSS. H_{m0} estimates produced via empirical gain approach are quite close to those obtained by lidar: $R^2 \approx 0.92$, RMSE $\approx$ 0.16 m, and a fit slope of 0.83 ± 0.02 (95% CI 0.79–0.87), indicating strong agreement. Evidently, the frequency (and period) response is weakly sensitive to the choice of gain, whereas the total energy contained within the spectrum is quite sensitive to the gain. The values of T_p computed via E-PSS agree well with those obtained from the lidar spectra: $R^2 \approx 0.72$, RMSE $\approx$ 1.00 s, and a fit slope of 0.95 ± 0.02 (95% CI 0.90–0.99), close to but marginally below 1:1.

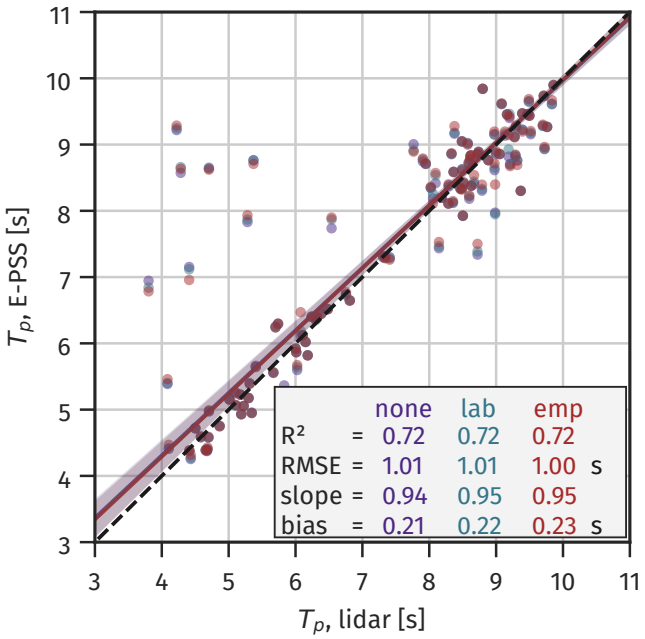


Fig. 8. Analogue to Fig. 7 for the peak wave period T_p of (2).

For the rest of the results and discussion contained within this manuscript, all E-PSS measurements presented were computed with the *empirical* DoLP gain.

C. Extending Across Multiple Scales

Although the long wave inversion results are promising, there remains a discontinuity between the scales directly resolved (frame size and smaller) and the scales for which wave spectra are inferred through the time domain ($f > 0.5$ Hz, or $\approx$ two full frame sizes). In order to bridge this gap, we performed a *time domain* long wave inversion in tandem with numerical integration of the slope fields to resolve the short wave elevation field. Between the two components, the short wave slope integration is at once both more computationally expensive and straightforward. We use the approach of Harker & O'Leary for

surface reconstruction from gradient fields [30]. Wavenumber-frequency directional spectra corresponding to waves directly resolved in the imager field of view are computed via direct three-dimensional FFT [9], so our surface reconstruction is only needed to bridge the intermediate scales between the direct spectrum and the long wave spectrum. Therefore, we are able to save some computational expense by reducing our spatial and temporal resolution before numerical integration: from 5.7 mm to 9.1 cm in space and from 30 Hz to 10 Hz in time. Slope field arrays were subjected to spatial Gaussian smoothing before block-averaging and temporal lowpass filtration before decimation in time in order to minimize the effect of aliasing. The time-domain reconstruction starts from the simple premise that one may infer surface elevation given a two-component slope time series, as in (1). An example of the finished product—the slope (s_E, s_N) to elevation (η_{long}) process—is given in Fig. 9a-b. Panel (c) shows the short (~ decimeter to meter scale) wave elevation field corresponding to a moment in time– specifically, at a node in the long wave elevation. The omnidirectional spectrum formed by summing individual slope spectra is, by construction, directionally complete. In order to ensure that the wave slope Fourier amplitudes are added in such a way to respect wave direction, we need to know the wave direction on a per-frequency basis.

More precisely, we need to define a single wave direction per frequency. Although a wavelet-based approach would allow for estimation of wave direction per frequency per time step, we opted for the simpler approach: Fourier transforms on the source slope time series, yielding one direction per frequency for each ten-minute segment. These directions (with 180° ambiguity resolved from the relative phase of the two orthogonal slope measurements) are used to construct a long wave phase function. We then "recolor" the elevation spectrum to take advantage of the directional completeness of (1). The first step in this process is to compute the wave amplitude spectrum $A(f)$ (real-valued, given in units of meters). This is assembled directly from the Fourier transforms ($\mathcal{F}\{\ldots\}$) of the individual slope component time series:

$$\begin{aligned} \hat{S}_E(f) &= \mathcal{F}\{s_E\, w\}/w_{\text{norm}} \\ \hat{S}_N(f) &= \mathcal{F}\{s_N\, w\}/w_{\text{norm}} \\ M(f) &= \sqrt{|\hat{S}_E|^2 + |\hat{S}_N|^2} \\ A(f) &= G(f)M(f)/k(f). \end{aligned} \tag{3}$$

Here, w is the tapering function (we chose a Tukey window with taper value $\alpha = 0.25$) and w_{norm} is the window-power debias $w_{\text{norm}} \equiv \sqrt{\langle w^2 \rangle}$). The frequency-dependent long wave direction θ_f may be extracted from the spectral amplitudes:

$$\begin{aligned} \cos\theta_f &= \frac{|\hat{S}_E|}{M} \\ \sin\theta_f &= \frac{|\hat{S}_N|}{M}\,\text{sign}\Big\{\text{Re}\big[\hat{S}_N \hat{S}_E^*\big]\Big\}. \end{aligned} \tag{4}$$

Here, sign{...} breaks ambiguity with help from the relative phase between $\hat{S}_E$ and $\hat{S}_N$. However, we don't need θ_f for its own sake; it is a pass-through that only serves to inform the relative balance of $\hat{S}_E$ and $\hat{S}_N$ in the construction of the elevation spectrum:

$$\varphi(f) = \arg\Big\{i\,\big[\cos\theta_f\,\hat{S}_E + \sin\theta_f\,\hat{S}_N\big]\Big\}. \tag{5}$$

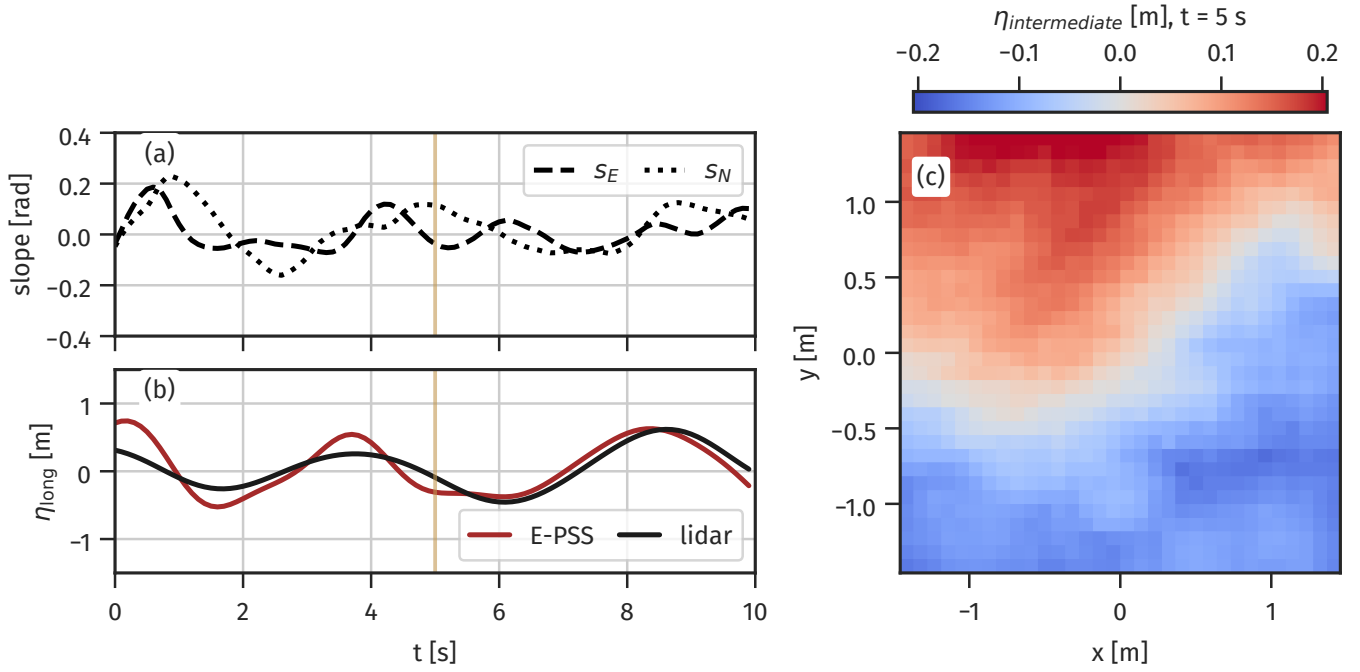


Fig. 9. Multiscale surface elevation inference. (a) long wave slope, (b) long wave surface elevation (with time-shifted lidar), (c) intermediate wave surface elevation computed from numerical integration of slope fields.

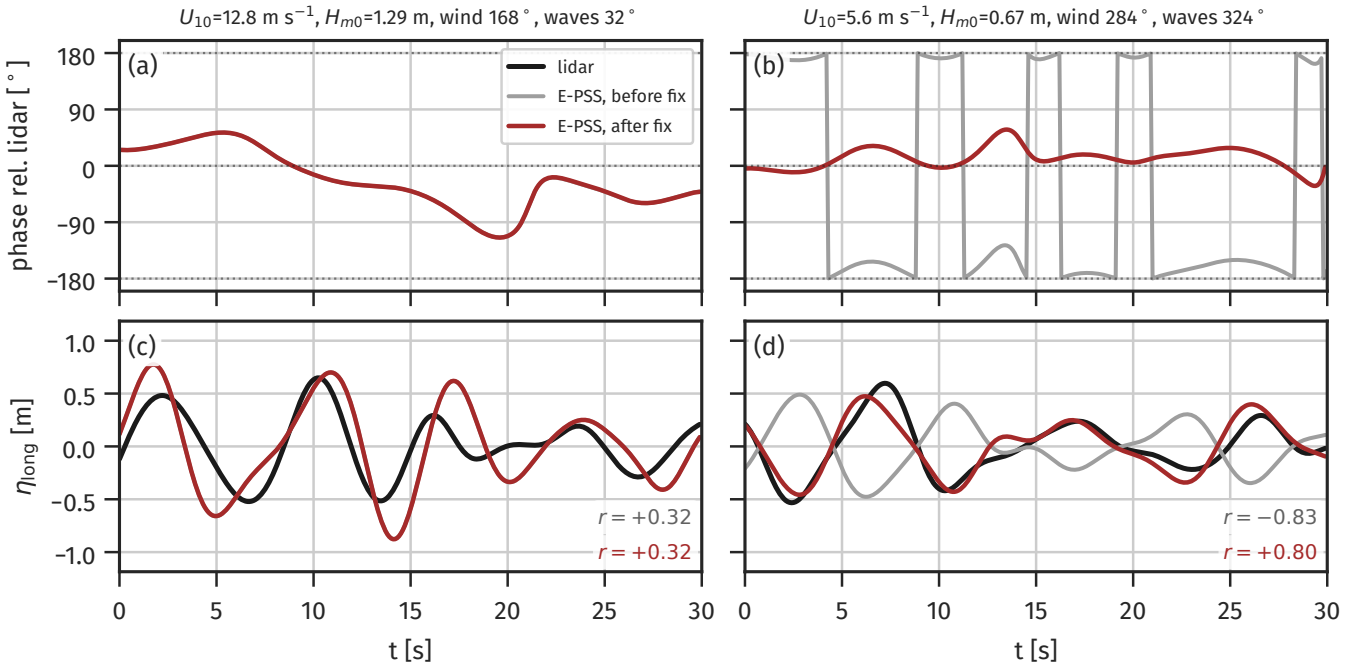


Fig. 10. Demonstration of need for curvature-based phase tiebreak in the inference of η_{long}. Top panels show E-PSS long wave phase relative to lidar, while bottom panels present surface elevation time series (before and after curvature-dependent ambiguity fix.

Equation (5) determines the phase of the surface elevation only to within 180°: $\cos\theta_f = |\hat{S}_E|/M$ is non-negative at every frequency by construction, so θ_f is confined to the eastern semi-circle. We resolve this ambiguity through interpretation of the surface *curvature* $\nabla^2\eta$ (exactly anti-phase with the surface elevation):

$$\nabla^2\eta(t) = \frac{\partial(\mathbf{s}\cdot\hat{\mathbf{c}})}{\partial c} + \frac{\partial(\mathbf{s}\cdot\hat{\mathbf{r}})}{\partial r}, \tag{6}$$

where $\hat{\mathbf{c}}$ and $\hat{\mathbf{r}}$ are the unit vectors of the image column and row axes (this is the penalty we must pay for rotating our slope measurements into earth coordinates). A hybrid (curvature and slope) approach to determining wave direction has been shown to provide improved fidelity relative to techniques which are restricted to wave slope only [31]. In practice, we compute $\nabla^2\eta$ on a per-frame basis via planar least-squares fit to the two-component slope field time series. Although knowledge of the surface curvature breaks the 180° tie, we cannot use its phase on a per-frequency basis: curvature scales as $k^2\eta$, and its computation over a 2.92 m frame will lead to noise-dominated phase information at swell frequencies. We instead use the original phase function $\varphi(f)$ as defined in (5), but allow the curvature phase to break ambiguity. This is done with a wave amplitude-weighted vote,

$$v(f) = \left|\mathcal{F}\{\nabla^2\eta(t)\}\right| \exp\left\{i\left[\arg\{-\mathcal{F}\{\nabla^2\eta(t)\}\} - \varphi(f)\right]\right\}. \tag{7}$$

In order to mitigate spurious jumps of 180° for regions with poor SNR, we smooth $v(f)$ over a bandwidth $\Delta f = 0.05$ Hz (yielding $\langle v\rangle$). The sign-resolved phase is then $\varphi(f) + 180° \times q(f)$, with the switch q thrown by the smoothed vote:

$$q(f) = \begin{cases} 1, & \mathrm{Re}\left[\langle v\rangle(f)\right] < 0 \\ 0, & \mathrm{Re}\left[\langle v\rangle(f)\right] \geq 0 \end{cases}. \tag{8}$$

Smoothing $v(f)$ in frequency does not degrade or erroneously broaden the spectrum. The true direction should not change ±180° over such a small frequency bandwidth, so this smoothing eliminates spurious jumps without negatively affecting measurement of realistic sea state veering. Furthermore, this smoothing helps to rescue the directional information at frequencies for which the curvature signal is too weak to break ambiguity. The final step is to construct the water surface elevation time series from the amplitude spectrum and the (updated, 360° unambiguous) phase function. In order to prevent amplification of noise at low frequencies which are poorly resolved by our small disc, we apply a highpass filter $h(f)$ with $f_c = 0.08$ Hz. This frequency corresponds to a wave approximately 50 × larger than our full framesize of 2.92 m (so, 100 × larger than our averaging disc). This combined inversion and filtration is performed in (9) below:

$$\eta_{\mathrm{long}}(t) = \mathcal{F}^{-1}\left\{h(f)A(f)\,e^{i\varphi(f)}\right\}. \tag{9}$$

The necessity of the curvature-dependent wave phase fix is demonstrated in Fig. 10. For waves traveling eastward ±90° with sufficiently high amplitude, the original slope phase of (5) tends to be sufficient to properly reconstruct the elevation time series (panels a,c). Waves with small amplitude at low frequencies (or which travel westward ±90°) are prone to 180° ambiguity (panels b,d). This correction has no bearing on the omnidirectional spectral energy densities or the integral parameters presented in the previous subsection (H_{m0} and T_p). However, correct determination of the instantaneous long wave phase is essential to the directional analysis which follows.

III. Results

A. Computing Wave Spectra

As demonstrated in section II (and shown in Fig. 6), the directionally-complete frequency spectrum obtained from the wave slope time series analysis exhibits good response for frequencies f between 0.08 Hz and 0.5 Hz. For the slope field size L^2 of 2.92 m × 2.92 m obtained during ASIT2019, this range of frequencies corresponds to waves with wavelength λ between $2L$ and $50L$. On the other end of the Polarimetric slope sensing is capable of obtaining the "direct" wavenumber-frequency directional spectrum for waves in the range of scales constrained by the sampling rate, camera resolution, and spatial frame size. For this field campaign, the sampling rate of 30 fps and effective spatial resolution of 5.69 mm/px limited the Nyquist frequency and wavenumber to 15 Hz and 551.74 rad m^{-1}, respectively. The imaging geometry imposes a hard constraint: the direct slope field Fourier analysis (and indeed, any direct elevation field Fourier analysis) suffers immensely in energy and direction fidelity at scales near the frame size.

In order to bridge the gap between the long wave information obtained strictly through the time domain (η_{long}) and the short wave information obtained through spatiotemporal analysis of waves directly resolved in-frame, we employed the extended wavelet directional method (EWDM) [32]. EWDM is a flexible framework for employing the wavelet directional method (WDM) of Donelan *et al.* (1996) [33] and other wavelet-based directional estimation techniques [34], [35].

The original WDM technique utilizes wavelet transforms of water surface elevation measurements made at spatially distributed (but closely clustered) gauges to directly compute the direction and wavenumber of each wave on a per frequency, *per time* basis. These may be integrated with respect to time to yield the elevation frequency and wavenumber-directional spectra $F(f,\theta)$ and $F(k,\theta)$, respectively. The technique has been used intermittently over the years [36], [37], [38], almost exclusively with measurements made from capacitance wave gauges spaced 0.10 m - 1.0 m. As discussed in section II-B, fixed spatial apertures are known to suffer at scales large and small relative to the aperture size (in this case, the maximum gauge separation, or *baseline*). In order to mitigate biases associated with aperture size, we employ the *MultiAperture* functionality of EWDM to construct clusters of virtual wave gauges over a range of aperture sizes, with the largest aperture ($A0$) the size of the frame and the smallest aperture (An) as compact as possible. This concept is demonstrated in Fig. 11, with panels (a-d) showing four select apertures overlaid on the short/intermediate water surface elevation field obtained via numerical integration of the coarse slope fields [30]. Panel (e) demonstrates the end result of this process: a multi-estimate stitch in which spectra computed from within each aperture are trusted in the center of their wavenumber range, with fringes gradually excluded via Gaussian weighting. The dark red curve represents the stitched (multi-aperture) spectrum $B(k) \equiv k^3F(k)$ while the black dashed curve represents the corresponding wavenumber spectrum obtained from direct FFT of the spatial slope fields ($B(k) \equiv k^2S(k)$).

Although our imaging-based technique offers a great deal of flexibility with respect to generating arbitrarily many virtual gauges, the total elevation field $\eta(x,y,t) = \eta_{\text{long}}(t) + \eta_{\text{intermediate}}(x,y,t)$ is artificial. The addition of $\eta_{\text{long}}(t)$ to all virtual wave gauges creates a "piston" of sorts, which reduces the phase difference between virtual gauges, therefore biasing the wavenumber estimate low. We mitigate this by performing the wavevector solve step in EWDM with a surface elevation field for which $\eta_{\text{long}}(t)$ has had contributions $f_c \approx 0.6$ Hz (so, 1.5 × frame size) removed. This partial lowpass filter effectively blunts the "piston" in frequency and phase space.

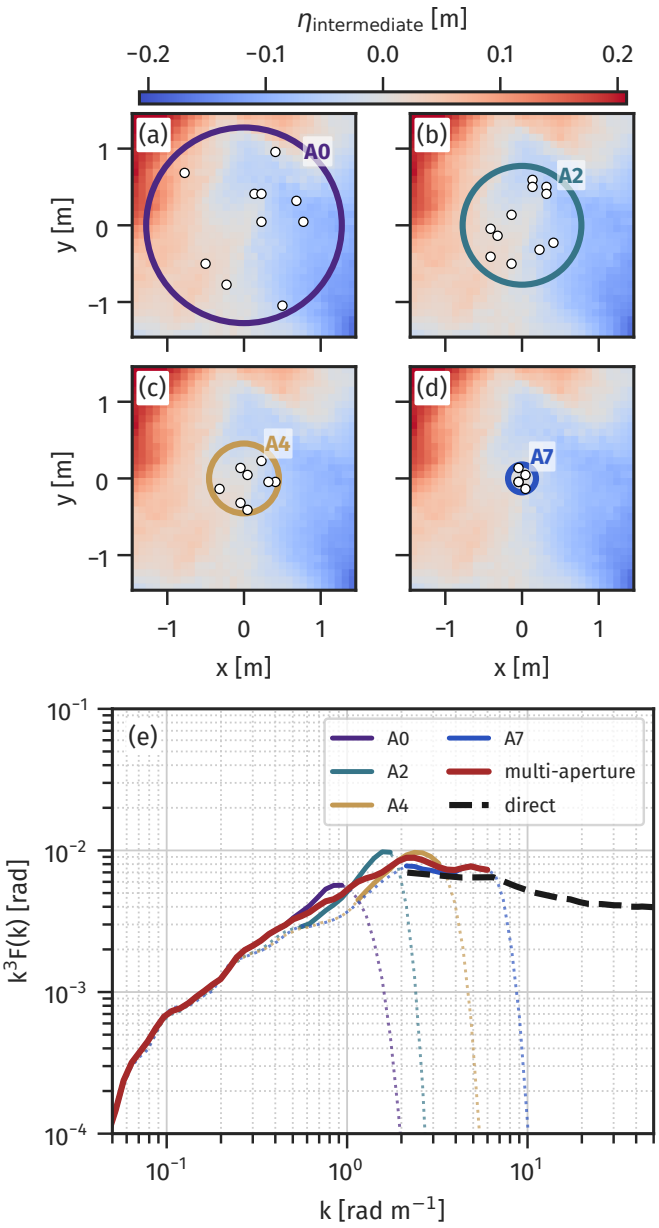


Fig. 11. Demonstration of multi-aperture EWDM processing. (a-d) example $\eta_{\text{intermediate}}(x,y)$ at a particular moment in time, with range rings corresponding to four different apertures enclosing virtual wave gauge clusters. (e) wavenumber omnidirectional saturation spectra produced from each individual aperture estimate (colors match aperture rings) and the composite (dark red) alongside the direct wavenumber saturation spectrum (dashed black).

The omnidirectional wave spectra obtained via E-PSS during ASIT2019 are shown in Fig. 12. Spectra have been averaged in wind speed bins of 2 m s^{-1} in U_{10} (as indicated by the color bar). The top row contains elevation spectra while the bottom row contains saturation spectra (i.e., compensated by $(2\pi f)^5/g^2$ or k^3). EWDM spectra are faded outside of a trusted band (vertical dashed gray lines), with the upper limit marking the handoff between EWDM and direct spectra (scales just larger/longer than those directly resolved in-frame). Dashed and dotted line segments correspond to the canonical equilbrium and saturation range slopes, respectively. The dashed equilibrium range takes an arbitrary (but convenient for visualization) vertical level, while the vertical level of the saturation range as plotted was taken from the WDM-based field measurements of Björkqvist *et al.* (2019) [38].

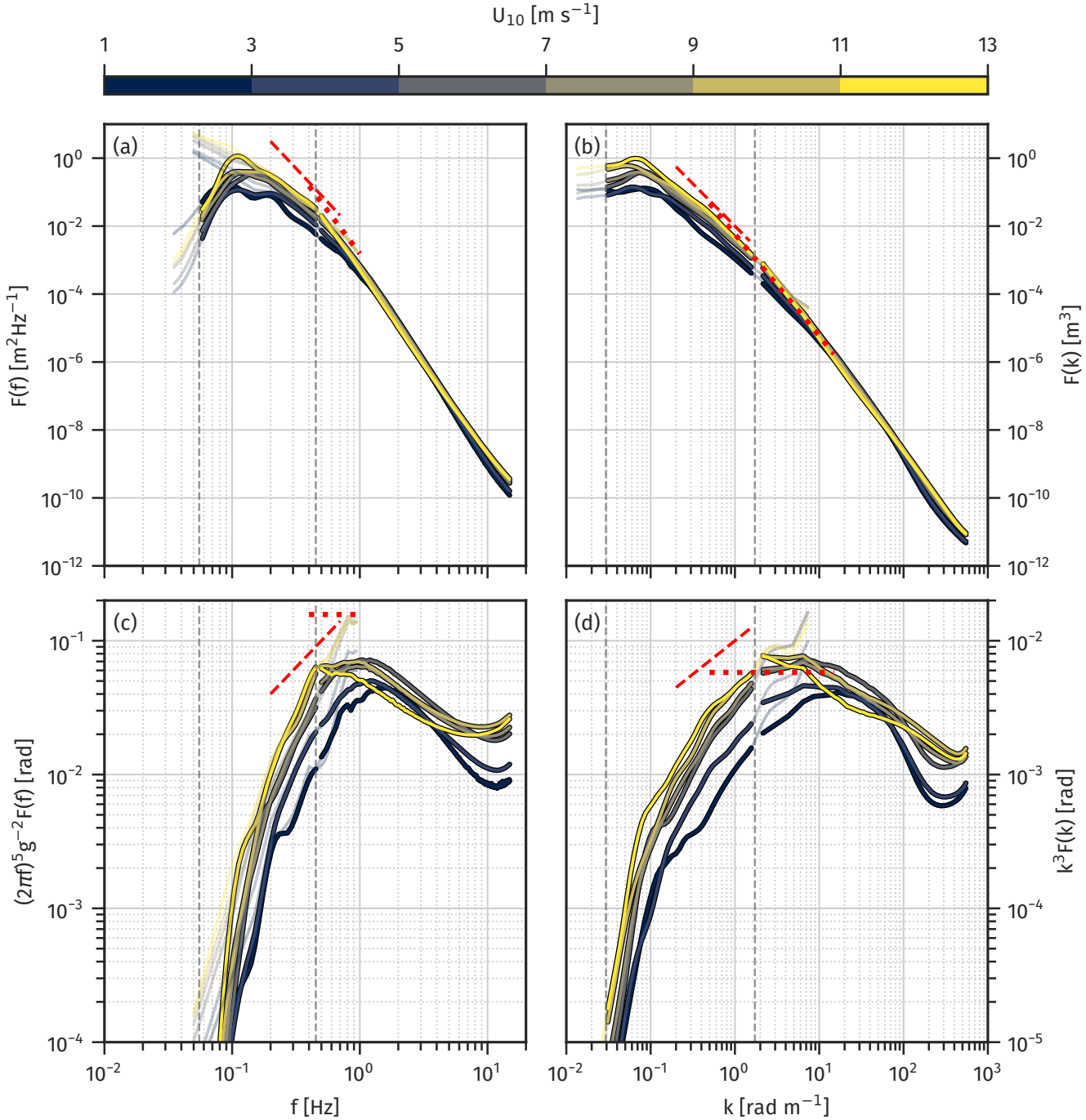


Fig. 12. Omnidirectional spectra: (a) frequency elevation, (b) wavenumber elevation, (c) frequency saturation, (d) wavenumber saturation. Color indicates the ten-meter wind speed range for which each bin average was computed. Dashed and dotted lines indicate the slope to be expected in the equilibrium (f^{-4}, $k^{-2.5}$ for the elevation spectrum; f^1, $k^{0.5}$ for the saturation spectrum) and saturation (f^{-5}, k^{-3} elevation; f^0, k^0 saturation) spectra, respectively.

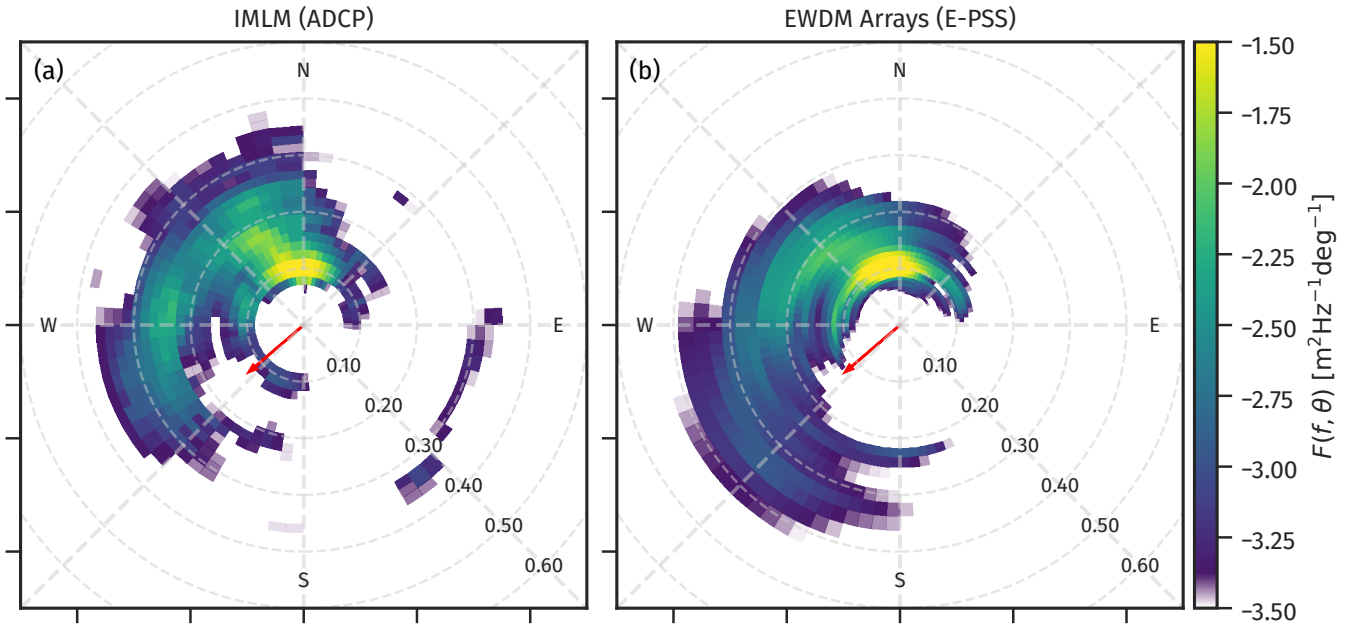


Fig. 13. Frequency-directional spectra computed given (a) the surface elevation/current profile measurements of the ADCP and (b) the measured two-component long wave slope and water surface elevation inferred via E-PSS. The red arrow indicates the wind direction (U_{10} = 7.7 m s^{-1}).

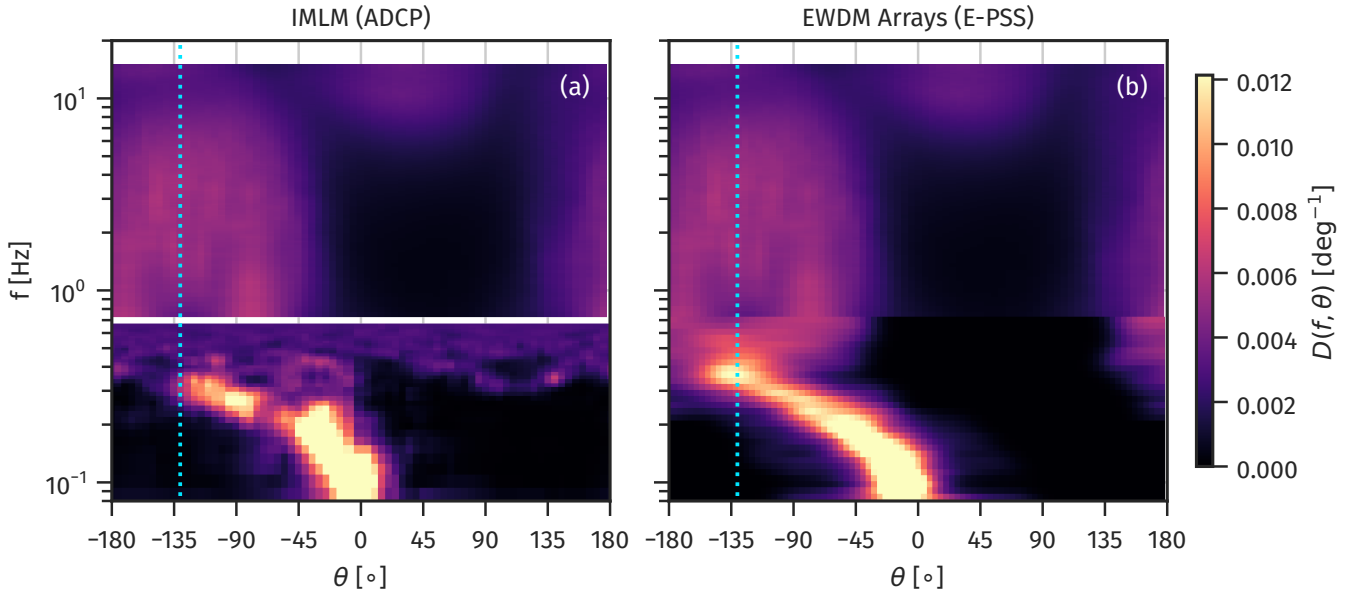


Fig. 14. Frequency-directional wave distribution $D(f,\theta)$ corresponding to the case shown in Fig. 13: (a) ADCP via IMLM, (b) E-PSS via EWDM Arrays, and (top half of a,b) directly from the wave slope field 3D Fourier transform. The cyan dotted line indicates the wind direction in "going-to" convention.

A pair of example frequency-directional wave spectra is shown in Fig. 13. The left panel contains the directional wave spectrum obtained from our Teledyne RDI Sentinel V ADCP [39], while the right panel contains the directional wave spectrum inferred from the surface elevation and slope measurements of E-PSS. The left spectrum was produced using an implementation of the Iterative Maximum Likelihood Method [40], [41] optimized for ADCP data and shipped as part of the standard Teledyne RDI analysis package. In IMLM, the maximum likelihood estimate is returned convolved with a steering-dependent smearing kernel, so that a lobe is always broader than the sea which produced it; passing each estimate back through the forward relation reproduces that kernel and makes the residual a correction which may be applied iteratively [42], [43]. The right spectrum was produced using EWDM Arrays (applied to our multi-aperture analysis of the total elevation field $\eta(x,y,t)$, as described in section III-A). Color corresponds to the base 10 logarithm of the spectral energy density. The red arrow gives the direction of the wind. It is often useful to consider the directional properties of the surface wave field irrespective of the frequency-dependent energy density. Given the frequency-directional spectrum $F(f,\theta)$, the

wave directional distribution $D(f,\theta)$ is defined as:

$$D(f,\theta) \equiv \frac{F(f,\theta)}{F(f)}, \tag{10}$$

where $F(f) = \int_{-\pi}^{\pi} F(f,\theta)\mathrm{d}\theta$ is the omnidirectional frequency spectrum. Fig. 14 contains $D(f,\theta)$ computed from the frequency-directional spectra shown in Fig. 13. The top portion of panels (a-b) correspond to the frequency directional spectra obtained directly from the water surface slope field "stacks". The bottom portion of those panels correspond to the directional spectra obtained via either the ADCP (IMLM) or E-PSS (EWDM MultiAperture Arrays) measurement. Although we have used EWDM as a demonstration of the capability for E-PSS to infer the long wave directional spectrum, the technique's measurements of $\eta(x,y,t)$ permit the use of a broad range of directional estimators. In Fig. 15, we show an expanded range of directional distribution $D(f,\theta)$ estimates. Panels (a) and (b) follow directly from Fig. 14. Panel (c) corresponds to $D(f,\theta)$ computed via EWDM's *Triplets* mode, a wavelet-based directional estimator for heave/pitch/roll (or three-axis displacement) measurements [34], [35]. Panels (d-f) show $D(f,\theta)$ computed from three traditional, yet still quite modern and optimized directional estimation techniques, respectively: IMLM [41], Extended Maximum Entropy Principle [44], and Bayesian Directional Method [45]. All estimators but EWDM *Arrays* (panels c-f) were fed the surface elevation and two-component wave slope time series (averaged over a circle inscribed within the FOV and subjected to *jinc* correction in frequency space), mimicking a 3 m discus buoy.

Among the directional distributions shown in Fig. 15, there is no extreme outlier. Of course– absent an independent reference for the frequency-directional behavior (e.g., as might be obtained via wide FOV imaging), we have no means with which to establish a *best* estimator for our measurement location. We see that all six estimators (ingesting two distinct data sources) identify the veering of a wave system from onshore ($\theta \approx 0°$) at low frequencies to obliquely offshore ($\theta \approx 135°$) into the wind direction at intermediate frequencies. One clear takeaway from this example is that comparing directional spectra to one another is no simple task. There is no lossless measure that adequately communicates the variation of the directional spectrum with frequency (or wavenumber, as the case may be). In the interest of comparing the directional estimates between methods (e.g., as in Fig. 15) across dozens of observational cases, we have chosen to reduce the information to be considered to the mean wave direction and the directional spreading width. Following Lin *et al.* (2021) [46], we calculate mean wave direction θ_0

$$\theta_0(f) = \int_{-\pi}^{\pi} \theta D(f,\theta)\mathrm{d}\theta, \tag{11}$$

and spreading width σ_θ

$$\sigma_\theta(f) = \left[\int_{-\pi}^{\pi} \left(\theta - \theta_0\right)^2 D(f,\theta)\mathrm{d}\theta\right]^{1/2}. \tag{12}$$

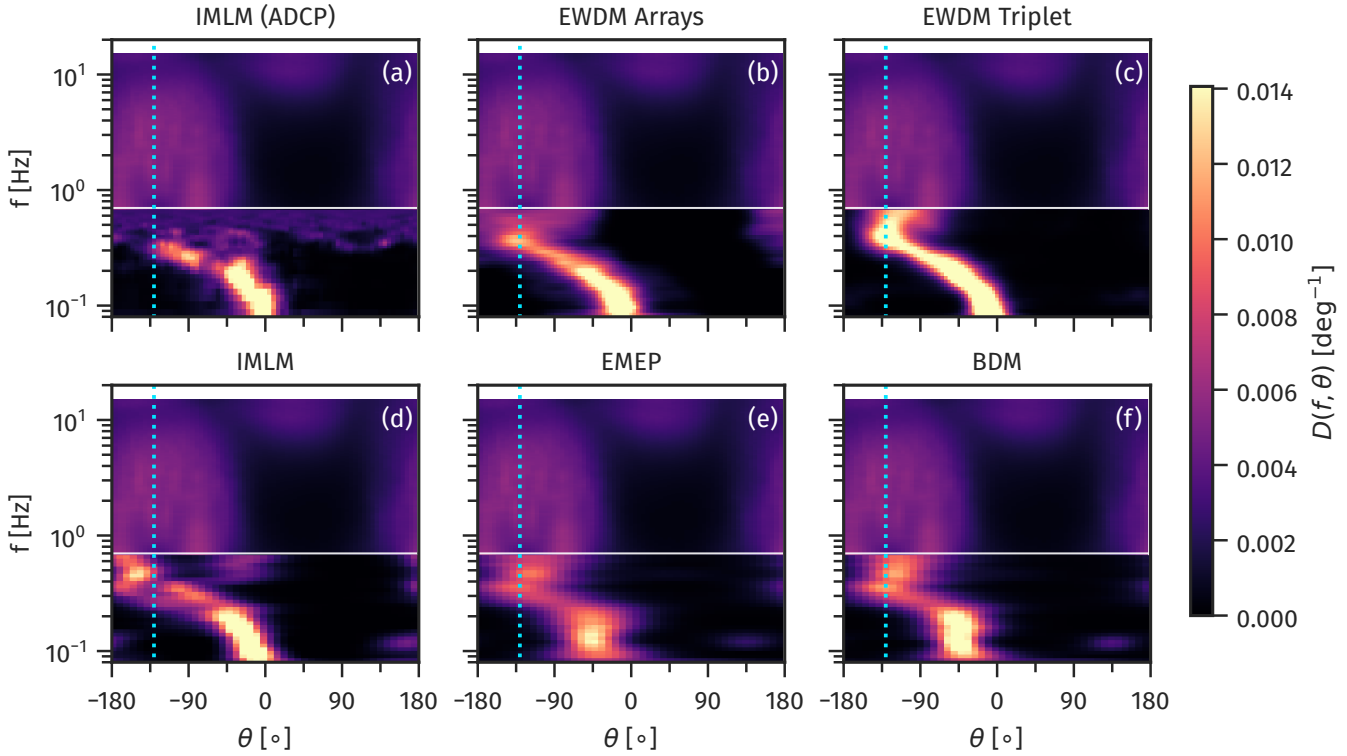


Fig. 15. Comparison of directional spreading function $D(f,\theta)$ computed from the ADCP via IMLM (a) with E-PSS directional estimators: (b) EWDM least squares wavevector solve (our default), (c) EWDM triplet with the elevation and two-component slope time series, (d) IMLM, (e) EMEP, (f) BDM. Wind direction (going-to convention) is given by cyan dotted line.

The mean wave direction and directional spread may alternatively be computed from the first-order Fourier moments [47]. We computed $\theta_0(f)$ and $\sigma_\theta(f)$ from the spectra themselves in order to maintain consistency between the ADCP and E-PSS estimates. This approach carries the additional benefit that it would also be usable with the "direct" frequency-directional spectra obtained from the wave slope fields themselves. In Fig. 16, we show the difference in mean wave direction $\Delta\theta_0$, with each θ_0 computed via (11) between the ADCP-derived and E-PSS-derived directional spectra. A direct intercomparison of mean wave direction estimates does strain our capability: there is a substantial spatial separation (≈1 km laterally) between the ADCP mooring and ASIT. This separation does allow for a good deal of evolution in the wave field given wind forcing, wave-current interaction, or breaking. We did attempt to correct for simple wave refraction via Snell's law, though this correction appears to have been negligible: the ADCP mooring was on the 18.3-meter isobath while ASIT is on the 15-meter isobath. Equation (12) provides the means to compute the wave directional spreading width $\sigma_\theta(f)$.

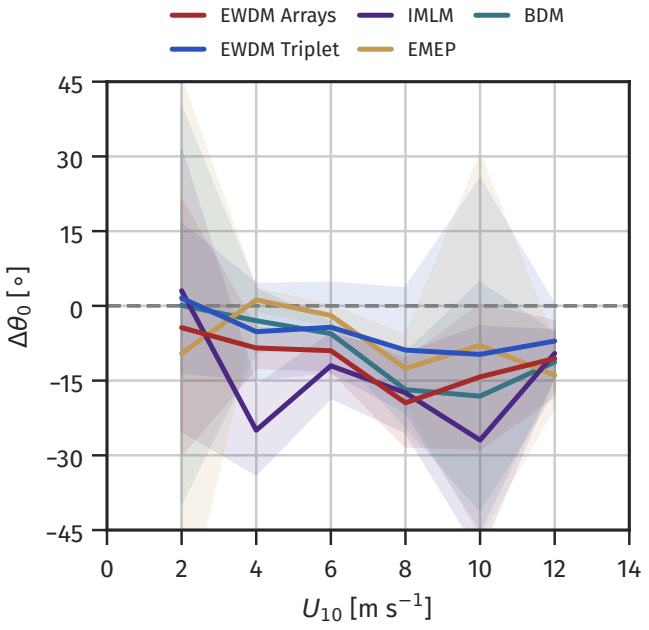


Fig. 16. Difference in mean wave direction θ_0 between E-PSS and the ADCP, bin-averaged in U_{10}, for each of the directional estimators of Fig. 15. Shaded bands mark the median absolute deviation within each wind speed bin.

In Fig. 17(a), we show an example $\sigma_\theta(f)$ computed from the spreading functions $D(f,\theta)$ obtained by ADCP and E-PSS for a particular run. The dotted black vertical segment corresponds to the peak frequency $f_p = 1/T_p$ of (2). In panel (b), we provide the bin averaged response of $\sigma_\theta(f_p)$ as a function of 10 meter wind speed U_{10}. The EWDM *Arrays* and *Triplets* spreading widths are consistent despite the fundamentally different means that each employs to obtain the wave direction [33], [34], [32]. We note that the apparent weak narrowing trend with increasing U_{10} is a consequence of the development of the sea state (and perhaps therefore a decrease in f_p). Our focus is on the intercomparison of ADCP and E-PSS-obtained directional spreading; we leave further discussion of wind-wave development to another paper. Also beyond the scope of the present study is an intercomparison of the directional estimators themselves in light of advances in spatiotemporal measurement of the wave field (e.g., revisiting Benoit's comprehensive study [48]).

IV. Discussion

A. Core findings of this paper

The results shown in Fig. 7 (H_{m0}) and Fig. 4 (mss) demonstrate that it is essential to characterize the polarization response of one's polarimetric camera– and that failure to properly account for ambient environmental conditions may have a substantial negative impact on the measured slope magnitude. As shown in Fig. 2b, unpolarized sky-leaving radiance will elevate the lowest degree of linear polarization that will be observed from a downward-looking imaging polarimeter. However, after sun glint-contaminated cases were excluded from the 2019-2020 ASIT polarimeter measurements, the remaining runs did not exhibit an elevated DoLP floor. To be specific, we found frequent occurrence of low (<0.1) values of DoLP for all remaining cases, indicating that variability in sky-leaving polarization state is likely to be of secondary importance once primary QC has been performed. However, given the somewhat large values of empirical DoLP gain (1.2-1.7) used during our field observations, upwelling radiance should not be neglected. This is consistent with previous studies of upwelling radiance [49], [50]. In any case, an approach which allows for a dynamic gain (e.g., our empirical approach) is likely to have a significant advantage over the static laboratory-based calibration. Our simple, single-camera "empirical" correction to the observed DoLP allows E-PSS to provide a reasonable estimate of the significant wave height H_{m0}. A recent intercomparison [51] between directional wave spectral estimates provided by different types of wave buoy systems (e.g., NOAA 3DMG, HIPPY, Triaxys, Watchman, and Datawell DWR) showed that for H_{m0}, it is reasonable to expect biases of approximately 0.1 m and RMSE of approximately 0.3 m. These values are in fact larger than the corresponding values we find for E-PSS relative to our reference lidar system, despite the enormous disparity in sample size between the two studies (order 10^4 for the buoy study, order 10^2 here). The buoy intercomparison of mean wave period T_{m02} is far more tightly clustered than our E-PSS-versus-lidar comparison, as expected given the disparity in sample size; we note that the buoy study reported T_{m02} whereas we report the peak period T_p, so that particular comparison is indicative rather than like-for-like.

The empirical gain approach appears to improve the estimate of mean square slope (Fig. 4), though our observed values are still lower than classic observations [23], [22] for moderate wind speeds ($U_{10} \geq 7$ m s^{-1}). It may be the case that these relatively low values of mss are owed to a degraded effective resolution of our polarimeter: for division of focal plane (DoFP) polarimeters like the one used in this study, one should expect a steep falloff in spectral energy density for scales 5-10 times larger than the image resolution [18]. We have computed the upwind and crosswind slope variance from the model

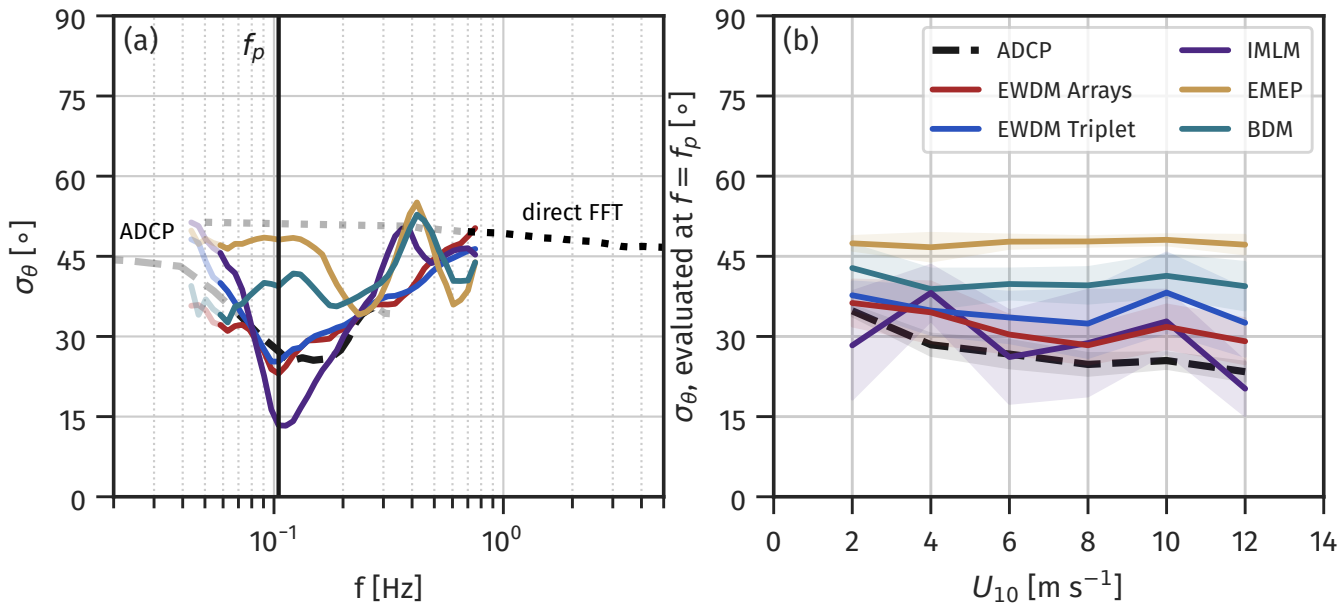


Fig. 17. Spreading width σ_θ, computed from $D(f,\theta)$ obtained from ADCP and E-PSS spectra following (12) for (a) a single case and given in terms of frequency and (b) all cases, bin-medians evaluated at the peak frequency f_p; shaded regions mark median absolute deviation.

wavenumber-directional spectrum of Elfouhaily *et al.* (1997) [19], setting the wavenumber upper limit of integration to 100 rad m^{-1}. It appears that this effective resolution limitation [18] is a reasonable explanation for the relatively low wave slope variance observed here. We found that slopes with absolute value greater than 0.3 are extraordinarily rare at moderate wind forcing, even with the empirical DoLP gain. Waves of such extreme slopes tend to be exceptionally short– e.g., sub-centimeter, in the short gravity-capillary and pure capillary regime. This low level of detection of ultra-steep waves is further evidence that the shortest waves are systematically underresolved [18]. In contrast, the sun glint-based statistical results of Cox & Munk and Bréon & Henriot capture the total effect of surface waves to contribute to reflectance– though those results do not provide insight into the scale dependence of the reflectance. It is this scale awareness that makes E-PSS so valuable.

Higher-order statistics (skewness and kurtosis) were estimated from the coefficients inferred through a least-squares fit of a Gram-Charlier expansion to the 2D Gaussian wave slope distribution; these are provided in Fig. 5. The effect of the DoLP gain on the computed wave slope skewness and kurtosis is inconclusive; it may be that the modest binned sample size of the present dataset (Fig. 5) is insufficient to the task of parsing the behavior of these higher-order statistics given adjustments to the measured light polarization state. Indeed, there is significant uncertainty in those parameters even as obtained from the large sample size satellite dataset used by Bréon & Henriot (2006). Nevertheless, the skewness and kurtosis values produced from the ASIT2019 field campaign are generally comparable to the classically-accepted values.

The combined omnidirectional spectra provided in Fig. 12 demonstrate the dual capabilities of E-PSS in providing a reliable measurement of both short waves and long waves. The polarization measurements used to compute the slope fields from which the spectra were produced were subjected to the empirically-derived DoLP gain. However, no further adjustments to the spectra were performed. The strong agreement in both spectral energy density and spectral subrange slope between the "direct" omnidirectional spectra and those inferred from the long wave time series is a natural consequence of the internal consistency of E-PSS. We sought to examine the ability of E-PSS to recover long wave directionality by computing the frequency-directional spectrum with a broad range of commonly (and uncommonly) used estimators.

The wave frequency directional spectra shown in Figs. 13, 14, & 15 provide a snapshot into the performance of E-PSS for capturing long wave directionality. Although it is difficult to say for certain without a third reference measurement, the smearing which occurs at the peak frequency in the ADCP-derived directional spectrum is likely a limitation of that particular measurement, whereas the directional veering we see in the E-PSS-derived spectra appears to track from the dominant long wave direction into the wind direction for shorter scales (Fig. 13). As shown in Fig. 14, this alignment of intermediate-to-short gravity wave scales into the wind direction is consistent between the IMLM-derived long wave and "direct" short wave portions of the E-PSS spectra. When considering mean/integral measures of wave directionality as obtained from the ADCP and E-PSS frequency-directional spectra, we found for mean wave directional difference $\Delta\theta_0$ the mean absolute error (MAE) to be 17.02° and the root-mean-square error (RMSE) to be 21.53°. The rather large RMSE computed from this comparison may result from the difference in frequency-dependent behavior that was observable via the ADCP and E-PSS (as mentioned in our discussion of the directional veering shown in Fig. 14). The directional spreading width at the peak frequency—i.e., $\sigma_\theta(f_p)$—determined via E-

PSS is within 10° of the ADCP-derived estimate on average (MAE = 5.89°, RMSE = 7.40°). This strong agreement is likely due to our narrow constraint that both ADCP and E-PSS estimates be considered at the same frequency, which reduces the scale-dependent variability which may have contributed to deviations in $\Delta\theta_0$. Although the locations and periods of operation are entirely different, this range of wave directional spread is consistent with values typically observed at the USACE FRF 8-m array [4], with $\sigma_\theta(f)$ ranging from 25° at 0.1-0.12 Hz to 45° at 0.25 Hz.

B. Impact on existing literature

The superior performance of the empirical DoLP correction (relative to no correction and a solely laboratory-based correction) is part blessing and part curse. On the one hand, it offers a way to ensure that existing datasets obtained with single surface-looking polarimeters may produce results which are robust to external environmental factors. However, it also indicates that previous applications of PSS—which did not take steps to account for variation in ambient illumination—may need to be reevaluated in light of the present results. Based on our review of existing literature on the topic (much of it contributed by the authors of this paper), we find a range of potential impacts. The studies most negatively impacted by a systematic underestimation of surface wave slope (and amplified case-to-case variability) are those with results based on the absolute spectral energy density or integrated slope variance at short gravity and gravity-capillary scales. These include field observations made with no laboratory calibration [52] and field observations made with a laboratory calibration [16], [9]. However, in each of these cases, only a subset of the results were tied to the absolute magnitude of the slope spectral energy density. Some key results were based on the analysis of relative wave response, e.g. hydrodynamic modulation [53] or suppression by surfactant slicks [17]. Many of the applications of PSS have bypassed this issue entirely, using the spatiotemporal propagation of the waves simply as a means to obtain the near-surface current [54], [55], [56], [57]; these results are weakly sensitive to the considerations mentioned here.

C. Going forward

In section I, we stated that the most complete form of PSS would entail simultaneous measurement of the sky-leaving, upwelling, and surface-reflected polarization state. Crucially, this would require an in-water sensor for obtaining measurements of upwelling radiance. The detector array used in our observations is the Sony IMX250MZR "Polarsens"; in recent years, this has become the standard detector for polarimetric cameras used for sensing ocean waves [18]. Per the manufacturer, the Sony IMX250 Pregius provides > 50% relative response between 400-800 nm light wavelengths. The spectral dependence of this response was not determined as part of the present study, though such a quantity is of import for techniques like PSSm [15] which depend on the wavelength dependence of DoLP; this separation allows one to infer properties about sky and surface-leaving radiance via Monte Carlo simululation [58], [59]. The rest of PSSm involves a series of nested iterations which culminate in (1) the recovery of the local relationship between DoLP and θ_i and (2) an isolated measure of the surface-reflected radiance. This technique shows a great deal of promise, and it may be the approach of choice for those who are able to make such multi-spectral polarization measurements (and then execute the multi-step iterative process).

Based on the results of the present work, we propose (without full demonstration of capability) a system that extends the empirical approach used in the present manuscript to include a richer base of information obtained by two surface-looking panchromatic polarimetric cameras: (1) one with a field of view (FOV) determined by the wave scales one desires to resolve and (2) one with a wide FOV, broad enough to capture angles of incidence ranging from near-nadir to past Brewster's angle (≈53° for an air-water interface). The in-frame angles of incidence θ_i would be determined from the camera-lens combination's intrinsic and extrinsic parameters (i.e., through simple geometry). The wide-FOV polarimeter would then measure the variation of DoLP with θ_i that is particular to a given measurement; this would inherently include all effects related to the intensity and polarization state of sky-leaving and upwelling radiance. This field-determined DoLP(θ_i) would then be used as a "lookup-table", allowing one to directly compute DoLP in the narrow-FOV scene without reliance on dedicated measurements or modeling of sky-leaving/upwelling radiance [58], [59] or any assumptions of theory. This approach is a bit cruder (and easier to employ) than PSSm, and is expected to be more robust than the narrow-FOV empirical approach we have used in the paper thus far. A demonstration of the sort of two-camera measurement which would enable this computation is provided in Fig. 18.

These observations were made on two non-consecutive days (9/29/2025 & 10/2/2025) from the end of the pier in Piermont, NY, USA (incidentally, the same location used for the field validation exercise of Zappa *et al.* (2008), [7]). On the overcast day (9/29/2025), an in-water turbidity sensor [60] reported river turbidity of 3.2 NTU (Nephelometric Turbidity Units); on the cloudless day (10/2/2025), the same sensor reported river turbidity of 0.3 NTU. It is well-documented that variation in turbidity/water type has a substantial impact on the intensity of upwelling radiance [61] and the polarization state of that upwelling radiance [62], [49]. Indeed, observations made on the day with overcast conditions and moderate water turbidity (Fig. 18a) are consistent with expectations that sky-leaving radiance be weakly polarized and upwelling radiance be minimal due to attenuation by in-water particulates. Conversely, observations made on the cloudless day (Fig. 18b) indicate significant impacts by sky-leaving polarization and upwelling radiance. Rather than discuss these complexities at greater length, we simply note the substantial variation in observed polarization state between the two days and reiterate the importance of taking this variation into account. For our demonstration, we made use of two azimuthally-aligned polarimetric cameras with identical Sony Polarsens DoFP polarimetric detectors. One camera was fitted with a narrow FOV lens (75 mm focal length, angles of view 6.5°×5.4°) and the other was fitted

with a wide FOV lens (5 mm focal length, angles of view 80.7°×70.7°). We found the mean DoLP measurements made by the two cameras to be comparable within their overlapping look region. Furthermore, in applying a least-squares fit to the data assuming a framework of specular reflection off the air-water interface, we were able to infer the Stokes vector of the sky-leaving radiance and the upwelling radiance fraction. It should be noted that the use of a wide-format lens may necessitate accounting for any effects resulting from light rays interacting obliquely with the polarizing filters on the focal plane array [63]. In any case, further testing of this two-camera concept is left to a future study.

We close this Discussion by noting that previous studies which made use of single-camera PSS [16], [52], [9] were performed with an understanding of the importance of diffuse, uniform illumination conditions. A great deal of observational cases were simply rejected during quality control; advancements like the ones we describe in this work will hopefully serve to allow for high quality wave measurements to be made in a broader range of conditions. The empirical gain

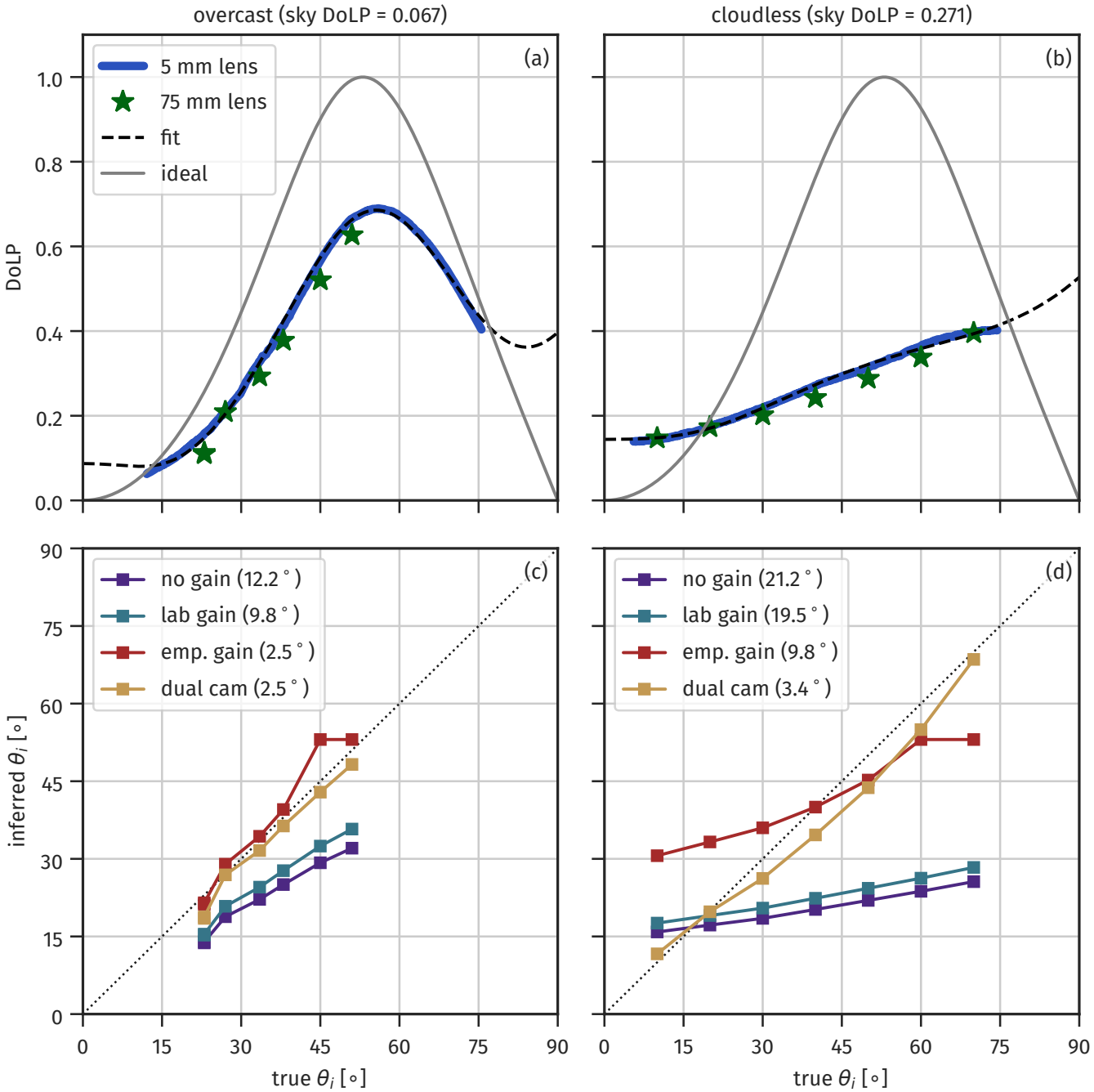


Fig. 18. Variation of DoLP with θ_i observed on days with overcast (left) and cloudless (right) sky conditions. Panels (a) and (b) show observed DoLP at each true θ_i (determined from camera look geometry). Panels (c) and (d) compare true incidence angle to incidence angle inferred via single-camera E-PSS and the novel dual-cam approach. Numerical values in legends represent mean absolute error. Column titles include sky DoLP observed directly by sky-oriented acquisition. On the overcast day, the least-squares fit determined $\hat{S}_{sky} = [1.000, 0.086, 0.016]$ (DoLP = 0.087) with an upwelling fraction of 0.000. On the cloudless day, the fit determined $\hat{S}_{sky} = [1.000, 0.117, 0.084]$ (DoLP = 0.144) with an upwelling fraction of 0.096.

approach we introduce in the present paper appears to be particularly well-suited towards mitigating the influence of upwelling unpolarized radiance, while the two-camera system concept we mention here (or PSSm [15]) is likely the optimal solution for making measurements in sunny conditions.

V. Summary

We have presented an extension of the polarimetric slope sensing (PSS) technique for measuring ocean surface waves. In the original implementation of PSS, we prefer to use a narrow field-of-view lens to avoid challenges associated with oblique ray interaction with the polarizing filters within the camera, though these effects may be mitigated through careful pre-processing [63]. The best measurements tend to be made with a camera incidence angle midway between nadir and Brewster's angle (i.e., $\approx 27°$); this helps to keep the particularly steep ocean wave surfaces away from camera-relative angles where the Fresnel curve is double-valued (i.e., $\approx 0°$ and $\approx 53°$). While it is essential that these effects be considered, accounting for each of them comes with a cost: (1) low camera incidence angles make the polarization measurement more susceptible to contamination by upwelling radiance (a consequence of the Fresnel equations); (2) the narrow field of view precludes direct measurement of intermediate to long-scale surface gravity waves. These challenges directly motivated the present work.

The technique we present here has been named "extended polarimetric slope sensing", or E-PSS; the "extended" modifier here denotes additional steps taken to

1) account for variation in ambient environmental conditions
2) allow one to infer the directional properties of surface gravity waves many times larger than the polarimeter FOV

When E-PSS is used with a simple empirical correction to the measured degree of linear polarization (DoLP), it is able to provide an accurate remote determination of the omnidirectional frequency spectrum across a wide range of surface wave frequencies (and therefore key sea state parameters, e.g. H_{m0}) from a single surface-looking imaging polarimeter. Further processing of the E-PSS outputs of the complete surface elevation $\eta(x, y, t)$ yields frequency-directional spectra which exhibit wave spreading comparable to the spectra obtained via a traditional ADCP-based approach. The mean direction for the shortest waves resolved through this indirect approach tends to be very close to value obtained from the short wave slope fields directly via multidimensional Fourier transform. We find consistency in spectral energy density and equilibrium range slope between the long wave and short wave components of the omnidirectional frequency spectra produced by E-PSS. Furthermore, the short wave statistics obtained directly from the wave slope fields after empirical DoLP correction exhibit stronger agreement with classic observations than has been shown with the unadjusted results of previous applications of PSS. Crucially, these measurements (both of long wave and short wave properties) are robust with respect to the ambient illumination condition. Finally, we present in concept an extension of this technique to involve two surface looking cameras, allowing for direct (in-scene) determination of the relationship between observed polarization state and surface inclination. This extension may improve performance in situations when the sky-leaving polarization state cannot be neglected, though that remains to be demonstrated at scale.

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The full suite of codes which contain our implementation of Polarimetric Slope Sensing (PSS) for DoFP polarimeters (as well as the additional modules which comprise E-PSS) are available at https://github.com/unh-cassll/polarimetric-slope-sensing. This repository contains a Jupyter notebook which demonstrates the start-to-finish process from raw DoFP image to slope field and inferred surface elevation time series. Wave spectra and slope statistics were computed using the framework available at https://github.com/unh-cassll/wave-slope-spectral-analysis. All of the codes used to produce the graphics shown in this document are available within the following self-contained GitHub repository: https://github.com/unh-cassll/E-PSS_paper. These codes make use of data products (wave spectra, slope/elevation timeseries, example graphics, etc.) stored on Zenodo (DOI: 10.5281/zenodo.17388349).

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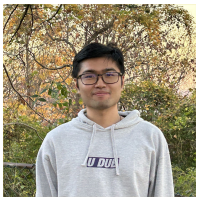
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